

IS 507 Group Project Proposal

From Farm to Fork: Assessing Economics Factors in Grain Consumption Choices

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Executive Summary

As students at Illinois, we are concerned about the volatility of corn prices, especially given that corn is the primary crop in the U.S. according to the USDA, and Illinois is one of the nation's leading corn-producing states. By understanding and predicting corn price fluctuations, we aim to assess their impact on farmers, food producers, and consumers like ourselves. In addition to corn, we are also interested in other crops, such as soybeans and barley, and plan to examine the price relationships among these crops. Our project will investigate the economic factors that influence grain consumption choices across various demographic and socioeconomic groups. Using machine learning techniques and statistical analysis, we will identify key economic drivers of grain consumption and develop predictive models to forecast changes under different economic conditions.

Objectives

1. Identify and quantify the impact of economic factors on corn production and consumption.
2. Develop predictive models to understand how changes in economic indicators affect grain demand.
3. Analyze the relationship between production and consumption choices in relation to supply and demand.
4. Provide insights for policymakers, agricultural planners, and stakeholders to enhance food security and economic resilience.

Research Questions

1. What are the primary economic factors influencing grain consumption choices?
2. How sensitive is grain consumption to changes in income, prices, and subsidies?
3. Can machine learning models accurately predict grain consumption trends based on economic indicators?

Methodology

The project will employ a combination of statistical analysis and machine learning techniques to assess and predict grain consumption based on economic factors.

1. Data Cleaning and Transformation:

- Standardize economic variables (e.g., price, income, inflation).
 - Normalize time-series data and create lag features to account for delayed effects of economic changes.
 - Encode categorical variables (e.g., subsidy presence) and handle missing values using imputation.
2. Model Development:
- Regression Models (e.g., Linear Regression, Ridge, Lasso) for quantitative predictions of grain consumption.
 - Tree-based Models (e.g., Random Forest, Gradient Boosting) to identify important economic predictors of consumption patterns.

Datasets and Sources of Information

1. USDA Economic Research Service (ERS): Historical data on grain production, consumption, and prices in the U.S. (USDA ERS).
2. FAOSTAT: Global agricultural data on production, imports, exports, and consumption by crop type. (FAO).
3. World Bank: Economic indicators such as GDP, inflation, and income per capita by country. (World Bank Open Data).
4. National Health and Nutrition Examination Survey (NHANES): U.S.-based survey data on demographic, socioeconomic, and dietary patterns (NHANES).

Potential Challenges and Mitigation Strategies

1. Data Availability: Comprehensive consumption and economic data might not be available for all regions. We will mitigate this by using proxies (e.g., regional data) or focusing on specific countries with robust data availability.
2. Data Quality and Completeness: Incomplete data may affect model accuracy. We'll handle missing data through imputation and validate the model using cross-validation techniques.
3. Model Interpretability: Complex models (e.g., Random Forest) can be challenging to interpret.